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Chapter 1: Governing Equations in Mantle Dynamics

1.1 Mechanics in Mantle Dynamics

Extensive studies have been conducted on fluid dynamics of a two-phase mixture. We briefly discuss the four governing equations of mechanics, drawing on a similar discussion presented by Drew and Segel (1971), Mckenzie (1984) and Katz (2022).

1.1.1 Phase Fraction

We begin with defining the phase fraction using an indicator function, under the assumption that multiple phases cannot coexist at the same spatial location. Accordingly, the phases are considered immiscible. The indicator function is defined as

$$\chi_i(\mathbf{x}) = \begin{cases} 1 & \text{phase } i, \\ 0 & \text{not phase } i. \end{cases} \tag{1.1}$$

We then define the macroscopic phase fraction function of phase i within a representative volume element V as

$$\phi_i = \frac{1}{V} \int_V \chi_i(\mathbf{x}) d\mathbf{v}, \tag{1.2}$$

where $\sum_i \phi_i = 1$. In the incompressible setting, the phase fraction may be equivalently regarded as the volume fraction occupied by each phase within a representative elementary volume, reflecting the proportion of space occupied by each phase at the macroscopic scale. Phase-averaged (or volume-averaged) quantities, also referred to as bulk quantities, are defined as

$$\overline{\gamma} = \sum_i \phi_i \gamma_i, \tag{1.3}$$

where γ_i denotes a quantity that may be scalar- or vector-valued.

1.1.2 Conservation Laws

We consider the conservation equation for a single-phase system. For a given volumetric quantity γ defined within a representative volume element V, we can write a corresponding conservation equation as:

$$\frac{d}{dt}\int_V \gamma d\mathbf{v} = -\int_{\partial V} \mathbf{f}\cdot d\sigma + \int_V S d\mathbf{v}, \tag{1.4}$$

where $\mathbf{f}$ represents flux travelling across the boundary of a representative volume element V. The minus sign indicates that the positive direction aligns with normal unit vector pointing outwards. The source/sink term S, represents the volumetric production rate of the volumetric quantity γ. We apply Gauss's theorem to obtain the differential form

$$\frac{\partial\gamma}{\partial t} + \nabla\cdot\mathbf{f} = S. \tag{1.5}$$

We model the partially molten mantle as a two-phase mixture consisting of liquid magma and solid mantle. The phase fraction of liquid magma is denoted by ϕ_l and phase fraction of solid mantle is labelled as ϕ_s, such that the relation $\phi_l + \phi_s = 1$ is always obtained.

We rewrite the single-phase conservation law (1.5) with the bulk density as

$$\frac{\partial\overline{\rho}}{\partial t} + \nabla\cdot\overline{\rho\mathbf{v}} = 0, \tag{1.6}$$

where $\overline{\rho} = \phi_l\rho_l + \phi_s\rho_s$. Under the Boussinesq approximation, we neglect the compressibility of each individual phase. Furthermore, we extend the Boussinesq approximation by disregarding the density difference between the two phases, except in the body force term. Accordingly, the continuity equation for the two-phase mixture can be written as

$$\nabla\cdot\overline{\mathbf{v}} = 0. \tag{1.7}$$

Similarly, the conservation equation for the phase-averaged momentum takes the form

$$\frac{\partial\overline{\rho\mathbf{v}}}{\partial t} + \nabla\cdot\overline{\rho\mathbf{v}\otimes\mathbf{v}} = \overline{\rho}\mathbf{g} + \nabla\cdot\overline{\boldsymbol{\tau}} - \nabla\overline{p}, \tag{1.8}$$

where $\boldsymbol{\tau}$ represents the deviatoric stress. By neglecting inertial effects, justified by the low Reynolds number of magma and mantle, the static force balance equation reduces to

$$\overline{\rho}\mathbf{g} + \nabla \cdot \overline{\boldsymbol{\sigma}} = 0, \tag{1.9}$$

where $\boldsymbol{\sigma} = \boldsymbol{\tau} - p\boldsymbol{I}$ represents the total stress, consisting of the deviatoric component $\boldsymbol{\tau}$ and the isotropic pressure $p\boldsymbol{I}$.

1.1.3 Darcy's Law

The equations governing bulk quantities (1.7) and (1.9) alone are insufficient to simultaneously determine the two distinct velocities and pressures. Therefore, an additional constitutive relationship is required to describe the behavior of the liquid phase within the porous medium. We examine more closely the interphase forces between the solid and liquid phases. A brief overview of the derivation of the individual governing equations for each phase is provided here, while more detailed and rigorous treatments can be found in the mixture theory literature, such as Bowen (1980). We write momentum balances for each individual phase as:

$$\begin{aligned} 0 &= \phi_l \rho_l \mathbf{g} + \nabla \cdot (\phi_l \boldsymbol{\tau}_l) - \nabla(\phi_l p_l) + \mathbf{F}_l, \\ 0 &= \phi_s \rho_s \mathbf{g} + \nabla \cdot (\phi_s \boldsymbol{\tau}_s) - \nabla(\phi_s p_s) - \mathbf{F}_s, \end{aligned} \tag{1.10}$$

where $\mathbf{F}_l = \mathbf{F}_s$ are the interphase forces between solid and liquid phases. The interphase forces between the solid and liquid phases are equal and opposite, in accordance with Newton's third law. Neglecting the deviatoric stress in the liquid phase and retaining only the isotropic pressure component, we balance the remaining stress with the viscous drag force and the equilibrium body force. Accordingly, we obtain the following relation for the relative velocity between phases, in alignment with Darcy's law

$$\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) = 0, \tag{1.11}$$

where k_{ϕ_l} is the matrix permeability, which usually takes the quadratic form as $k_{\phi_l} = k_0 \phi_l^2$.

1.1.4 Compaction

In the context of mantle dynamics, an additional yet unique and essential relationship regulates melt migration. We now revisit the continuous equation (1.6), and consider the two phases separately with the effects of melting incorporated, i.e. ,

$$\begin{aligned} \frac{\partial \phi_l}{\partial t} + \nabla \cdot (\phi_l \mathbf{v}_l) &= \Gamma, \\ \frac{\partial (1-\phi_l)}{\partial t} + \nabla \cdot ((1-\phi_l)\mathbf{v}_s) &= -\Gamma, \end{aligned} \tag{1.12}$$

where Γ is the unknown melting rate. We focus on the solid phase equation to get the evolving compaction relationship as

$$\frac{\partial \phi_l}{\partial t} + \mathbf{v}_s \cdot \nabla \phi_l = \Gamma + (1-\phi_l)\nabla \cdot \mathbf{v}_s, \tag{1.13}$$

where the divergence term $\nabla \cdot \mathbf{v}_s$ is often referred to as the compaction rate. We approximate the static compaction relation as

$$\xi_{\phi_l} \nabla \cdot \mathbf{v}_s = p_l - p_s, \tag{1.14}$$

where ξ_{ϕ_l} is often referred to as bulk viscosity or compaction viscosity. The form of the compaction viscosity ξ_{ϕ_l} has been the subject of extensive discussion. Mckenzie (1984) popularized the concept of compaction viscosity and relating the term to the stress balance in a partially molten system. Bercovici et al. (2001) extended this formulation by incorporating surface tension effects. Rabinowicz et al. (2002) further examined compaction behavior in the limit of infinitesimally small porosity. Following the approaches of Hewitt and Fowler (2008) and Katz (2008), we approximate the compaction viscosity ξ_{ϕ_l} using a simplified constitutive relationship,

$$\xi_{\phi_l} = \frac{\mu_s}{\phi_l}, \tag{1.15}$$

which highlights the fact that increased melting facilitates the compaction of the solid phase, while in the limit of no melting ($\phi_l = 0$), compaction is entirely eliminated.

1.1.5 Full Static System

By combining equations (1.7), (1.9), (1.11) and (1.14), We finally arive at the full static flow mechanical system,

$$\begin{aligned}
\phi_l(\mathbf{v}_l - \mathbf{v}_s) + \frac{k_{\phi_l}}{\mu_l}(\nabla p_l - \rho_f \mathbf{g}) &= 0, && (1.16)\\
\nabla \cdot (\phi_l \mathbf{v}_l + \phi_s \mathbf{v}_s) &= 0, && (1.17)\\
\nabla \bar{p} - \nabla \cdot \boldsymbol{\tau}(\mathbf{v}_s) + \bar{\rho}\mathbf{g} &= 0, && (1.18)\\
\mu_s \nabla \cdot \mathbf{v}_s - \phi_l(p_l - p_s) &= 0, && (1.19)
\end{aligned}$$

where the deviatoric stress of the solid is

$$\boldsymbol{\tau}(\mathbf{v}_s) = 2\mu_s\phi_s\big(\mathcal{D}\mathbf{v}_s - \tfrac{1}{3}\nabla \cdot \mathbf{v}_s \mathbf{I}\big), \tag{1.20}$$

and $\mathcal{D}\mathbf{v}_s = \frac{1}{2}(\nabla \mathbf{v}_s + \nabla \mathbf{v}_s^T)$ is the symmetric gradient. Moreover, we adjust the relative permeability $k_{\phi_l} = k_0 \phi_l^{2+2\Theta}$ with constant $\Theta \in [0, 1/2]$. Here μ_s and μ_l are dynamic viscosities for solid and liquid respectively. This complete static mechanical system relates the velocities and pressures of the solid and liquid phases to a given porosity distribution, thereby laying the foundation for subsequent computations.

1.1.6 Scaled Formulation

The Darcy equation mathematically degenerates in the regions where the porosity is zero. To handle this issue, we use the formulation developed previously by Arbogast et al. (2017). We first define pressure potentials as

$$q_l = p_l - \rho_l g z \quad \text{and} \quad q_s = p_s - \rho_l g z, \tag{1.21}$$

where $g = |\mathbf{g}|$ is the magnitude of gravity acceleration and z is the depth, such that $\mathbf{g} = g\nabla z$. Note that ρ_l is used to define both potentials, so that with $q = \bar{q} = \bar{p} - \rho_l g z$,

$$p_l - p_s = q_l - q_s = \frac{1}{1-\phi_l}(q_l - q).$$

In the mechanical system, the subscript on ϕ_l will be omitted. and we will simply write it as ϕ. Accordingly, the solid phase fraction will be expressed as $1-\phi$, emphasizing the constraint that the two phase fractions sum to one. Thus

$$\begin{aligned}
\phi \mathbf{v}_r + \frac{k_0 \phi^{2+2\Theta}}{\mu_l} \nabla q_l &= 0, && (1.22)\\
\mu_s \nabla \cdot (\phi \mathbf{v}_r) + \frac{\phi}{1-\phi}(q_l - q) &= 0, && (1.23)\\
\nabla q - \nabla \cdot \tau(\mathbf{v}_s) &= -(1-\phi)\rho_r \mathbf{g}, && (1.24)\\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi}{1-\phi}(q_l - q) &= 0, && (1.25)
\end{aligned}$$

where the relative velocity $\mathbf{v}_r = \mathbf{v}_l - \mathbf{v}_s$ and the density difference $\rho_r = \rho_s - \rho_l$. Scaled variables are also defined,

$$\tilde{\mathbf{v}}_r = \phi^{-\Theta} \mathbf{v}_r \quad \text{and} \quad \tilde{q}_l = \phi^{1/2} q_l, \tag{1.26}$$

and the problem is reformulated as

$$\begin{aligned}
\tilde{\mathbf{v}}_r + \frac{k_0 \phi^{1+\Theta}}{\mu_f} \nabla(\phi^{-1/2} \tilde{q}_l) &= 0, && (1.27)\\
\mu_s \phi^{-1/2} \nabla \cdot (\phi^{1+\Theta} \tilde{\mathbf{v}}_r) + \frac{1}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) &= 0, && (1.28)\\
\nabla q - \nabla \cdot \tau(\mathbf{v}_s) &= -(1-\phi)\rho_r \mathbf{g}, && (1.29)\\
\mu_s \nabla \cdot \mathbf{v}_s - \frac{\phi^{1/2}}{1-\phi}(\tilde{q}_l - \phi^{1/2} q) &= 0, && (1.30)
\end{aligned}$$

where the new scaled relative velocity maintains numerical stability in the zero prosority regions.

1.2 Transport of Chemical Species and Energy

With the velocities of solid and liquid phases determined, we are able to evolve our system forward in time. In our system, chemical species and energy are transported by the motion of the solid and the liquid phases.

1.2.1 Transport of Chemical Species

The realistic chemical compositions of mantle and magma are highly complicated and very hard to trace at the same time. Therefore, we present a simple two-species two-phase model. In this framework, we also negelect detailed description of the chemical reaction process and exclude the effect of the change of Gibbs free energies. Similar discussion can be found in the literature like Spiegelman and Elliott (1993), Hewitt and Fowler (2008), Katz (2008) and Hesse et al. (2022). Both species can exist in either the solid or liquid phase. They are assumed to be immiscible in the solid state but fully miscible in the liquid phase. By again extending the Boussinesq approximation, we consider a homogenous density ρ for both the solid phase and the liquid phase. Figure 1.1 illustrates the coexistence of species and phases in representative volume element. Within the representative volume element V, $\gamma_{i,j}$ represents

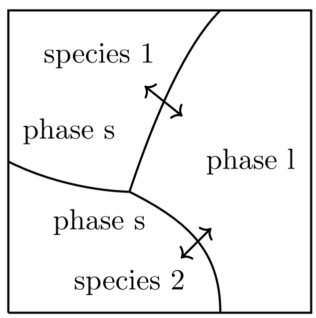

Figure 1.1: Phases and species in an representative volume element (RVE).

one quantity of the species i present in phase j, where $i \in 1, 2$ and $j \in \{s, l\}$. The mass fraction $c_{i,j}$ of a given species i in phase j is also regarded as a concentration. We can write transport equations for an individual species i, assuming a pseudo melting term Γ representing the transition between the solid and the liquid phase as

$$\frac{\partial}{\partial t}\big(\phi_l \rho_l c_{i,l}\big) + \nabla \cdot \big(\phi_l \rho_l c_{i,l} \mathbf{v}_l - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\big) = \Gamma, \tag{1.31}$$

$$\frac{\partial}{\partial t}\big(\phi_s \rho_s c_{i,s}\big) + \nabla \cdot \big(\phi_s \rho_l c_{i,s} \mathbf{v}_s - \phi_s \rho_s \mathcal{D}_{i,s} \nabla c_{i,s}\big) = -\Gamma, \tag{1.32}$$

where $\mathcal{D}_{i,l}$ and $\mathcal{D}_{i,s}$ are the diffusion coefficients of component i in the liquid and the solid phase respectively. The diffusion term will be dropped in the solid phase. By summing up equations (1.31) and (1.32), we obtain an advection-diffusion equation as

$$\frac{\partial \overline{\rho c_i}}{\partial t} + \nabla \cdot \left(\overline{\rho c_i \mathbf{v}} - \phi_l \rho_l \mathcal{D}_{i,l} \nabla c_{i,l}\right) = 0, \tag{1.33}$$

where the phase averaged quantities are $\overline{\rho c_i} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j}$ and $\overline{\rho c_i \mathbf{v}} = \sum_{j\in\{s,l\}} \phi_j \rho_j c_{i,j} \mathbf{v}_j$. The subscript i will be omitted, as only one chemical species needs to be explicitly computed in this two-species system. By extending the Boussinesq approximation such that $\rho_l = \rho_s = \overline{\rho}$, the species transport equation can be simplified to the form

$$\frac{\partial \overline{c}}{\partial t} + \nabla \cdot \left(\overline{c\mathbf{v}} - \phi_l \mathcal{D}_l \nabla c_l\right) = 0. \tag{1.34}$$

1.2.2 Effective Velocity

The concept of effective velocity has long been employed in chemistry, particularly in the context of particle transport. In mantle dynamics, Spiegelman (1996) applied effective velocity for equilibrium transport of concentration within the liquid phases. We define a corresponding effective velocity for the transport of bulk concentration. To begin, we define the partition coefficient as $D_i = c_{i,s}/c_{i,f}$. The concentration of species i in the liquid and solid phase can be expressed in terms of the bulk concentration as follows:

$$(\phi_l + \phi_s D_i) c_{i,l} = \overline{c}_i, \tag{1.35}$$

$$(\phi_l / D_i + \phi_s) c_{i,s} = \overline{c}_i. \tag{1.36}$$

By expressing $c_{i,l}$ and $c_{i,s}$ with bulk concentration given by equations (1.35) and (1.36), we can rewrite the bulk velocity as

$$\overline{c_i \mathbf{v}} = \phi_s \mathbf{v}_s c_{i,s} + \phi_l \mathbf{v}_l c_{i,l} = \left(\frac{\phi_l \mathbf{v}_l + D\phi_s \mathbf{v}_s}{\phi_l + \phi_s D}\right) \overline{c}_i. \tag{1.37}$$

We get the proper expression for the effective velocity, denoted as

$$\mathbf{v}_{\text{eff}} = \frac{\phi_l \mathbf{v}_l + D_i \phi_s \mathbf{v}_s}{\phi_l + \phi_s D_i}, \tag{1.38}$$

which is the appropriate velocity at which the bulk concentration should advect. In the absence of melting, where $\phi_l = 0$, we retrieve $\mathbf{v}_{\text{eff}} = \mathbf{v}_s$. Conversely, in the case of complete melting, where $\phi_l = 1$, we obtain $\mathbf{v}_{\text{eff}} = \mathbf{v}_l$. Therefore, the definition of effective velocity is well defined in those two limiting scenarios. We rewrite the transport equation (1.34) with effective velocity (1.38) as

$$\frac{\partial C}{\partial t} = \nabla \cdot (\mathbf{v}_{\text{eff}} C - \phi_l \mathcal{D}_l \nabla c_l) = 0, \tag{1.39}$$

where $C = \overline{c}$ for simplicity.

1.2.3 Transport of Energy

According to the first law of thermodynamics, we write the energy balance equation in the phase-averaged internal energy form as

$$\frac{\partial \overline{\rho u}}{\partial t} + \nabla \cdot \overline{\rho u \mathbf{v}} = \overline{\rho \mathcal{Q}} + \nabla \cdot \overline{K} \nabla T + \nabla \cdot \overline{\boldsymbol{\sigma} \cdot \mathbf{v}} + \overline{\rho \mathbf{v}} \cdot \mathbf{g}, \tag{1.40}$$

where $\overline{K} = \phi_l K_l + \phi_s K_s$ is the phase averaged thermal conductivity. On the left-hand side, the equation represents the time-dependent rate of change of internal energy of a two-phase mixture as it moves with the flow. On the right-hand side, it includes the heat source term $\mathcal{Q}$, conductive heat transfer following Fourier's law, mechanical work done by mixture stress, and work against gravity respectively.

In a thermochemical equilibrium scenario, we replace internal energy u with enthalpy h. The specific enthalpy is defined as

$$h = u + PV/\rho, \tag{1.41}$$

We formulate the enthalpy equation by neglecting the external heating/cooling term $\mathcal{Q}$ and dissipative heating following the discussion presented by Katz (2008),

$$\frac{\partial \overline{\rho h}}{\partial t} + \nabla \cdot \overline{\rho h \mathbf{v}} = \frac{\partial P}{\partial t} + \overline{v} \cdot \nabla P + \nabla \cdot \overline{K} \nabla T. \tag{1.42}$$

1.2.4 Temperature Formulation

The specific enthalpy relates to temperature directly with specific heat capacity as

$$h - h_0 = c_p(T - T_0), \tag{1.43}$$

where the reference values h_0 and T_0 are set as zero for simplicity. Another important physical attribute in the context of melting is latent heat L. We write specific enthalpy in the solid and the liquid phase separately as

$$\begin{aligned} h_s &= c_p T, \\ h_l &= c_p T + L. \end{aligned} \tag{1.44}$$

We rewrite the enthalpy equation with latent heat and temperature as

$$\frac{\partial \overline{\rho h}}{\partial t} + \rho c_p \nabla \cdot (\overline{\mathbf{v}} T) = \rho L \nabla \cdot (\phi_s \mathbf{v}_s) + \overline{K} \nabla^2 T + \rho \alpha_\rho T \overline{\mathbf{v}} \cdot \mathbf{g}, \tag{1.45}$$

where one specific heat capacity c_p is used for both components in solid and liquid phase. The last term including gravitational acceleration has previously been referred to as adiabatic cooling or decompression cooling by Mckenzie and Bickle (1988). Assuming no density difference between the solid and the liquid phase, we divide ρc_p on both sides of the equation to obtain a normalized form as

$$\frac{\partial \overline{H}}{\partial t} + \nabla \cdot (\overline{\mathbf{v}} T) = \frac{L}{c_p} \nabla \cdot (\phi_s \mathbf{v}_s) + \kappa \nabla^2 T + \frac{\alpha_\rho}{c_p} T \overline{\mathbf{v}} \cdot \mathbf{g}, \tag{1.46}$$

where $\overline{H} = \overline{\rho h}/\rho c_p = \overline{h}/c_p$ is the bulk specific enthalpy and $\kappa = \overline{K}/\rho c_p$ is the bulk heat diffusivity.

1.3 Eutectic Phase Behavior

With the transport equations defined, we can now proceed to redistribute chemical species and energy within the system. The next step is to determine the thermodynamic state of the system, assuming thermodynamic equilibrium is obtained at each time step. Recognizing that mid-ocean ridges form in regions where temperatures are generally lower, eutectic melting, where the presence of a secondary species significantly lowers the melting temperature, provides a suitable model for describing the melting behavior in these areas.

The binary eutectic system has been widely applied in studies involving chemical equilibrium, with detailed description can be found in textbooks such as Lupis (1983). In the context of mantle dynamics, eutectic melting was being incorporated by Ribe (1985). Moreover, eutectic phase behavior has been applied to solidification of alloys by Wheeler et al. (1996). More recently, binary eutectic melting has also been applied to water-salt-brine systems in glaciers Hesse et al. (2022).

We assume the two primary components in our partially molten system are olivine and orthopyroxene(opx), which are highly miscible in the liquid phase, but immiscible in the solid phase. Olivine is considered the dominant component, while orthopyroxene represents the minor component. Assuming thermodynamic equilibrium is obtained at each time step, we can determine the phase fraction of liquid, the volume fraction of each solid components and the temperature using a binary eutectic phase diagram with transported enthalpy and bulk composition.

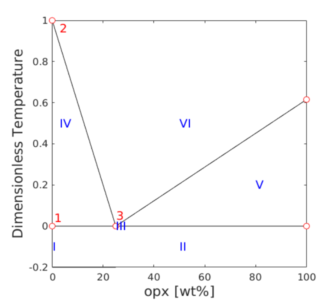

Figure 1.2: Eutectic phase diagram of a Olivine-Orthopyroxene system.

A typical eutectic phase diagram is shown in Figure 1.2. There are two key concepts about the diagram:

- Eutectic melting point or eutectic temperature T_e, is the lowest temperature a eutectic system starts melting. The melting happens at eutectic temperature is called eutectic melting. The temperature of the system is fixed as eutectic temperature when eutectic melting happens.

- Eutectic composition C_e is a fixed opx-olivine concentration ratio of the liquid produced by eutectic melting.

1.3.1 Clausius-Clapeyron Relation

In geological contexts, pressure can undergo significant variations during dynamic events such as mantle upwelling. As a result, the influence of pressure on the melting point cannot be neglected. To account for this, we approximate the solidus melting temperature T_m at a given pressure using the approximated Clausius–Clapeyron relationship. We also extend such approximation to eutectic temperature T_e as

$$\begin{aligned} T_m &= T_{m0} + \nu^{-1}(P - P_0), \\ T_e &= T_{e0} + \nu^{-1}(P - P_0), \end{aligned} \tag{1.47}$$

where ν is the Clausius-Clapeyron constant, and P_0 is the reference pressure usually set to be 0. The melting point T_{m0} and eutectic point T_{e0} refer to their respective values at standard atmospheric pressure.

1.3.2 Some Assumptions

In a eutectic system, the current thermodynamic state of the material can be identified using the bulk mass fraction of orthopyroxene (opx) and bulk enthalpy of the system. Our bulk mass fraction will refer to opx as we consider a binary system involving only two components. We begin by introducing some assumptions regarding densities and other material properties.

- We assume no density difference between solid phase olivine and opx, i.e., $\rho_{\text{opx}} = \rho_{\text{olivine}}$.

- We assume no density change during phase transition (extended Boussinesq approximation), i.e., $\rho_l = \rho_s$.
- We assume constant and uniform material thermodynamics properties like specific heat capacity c_p and thermal expansion coefficient α_p.

Following the assumptions, we relate the bulk composition with previously defined phase averaged concentration (1.39) as

$$\frac{\phi_l \rho_l c_l + \phi_s \rho_s c_s}{\phi_l \rho_l + \phi_s \rho_s} = \phi_l c_l + \phi_s c_s = \phi_2 + \phi_l c_l = C, \tag{1.48}$$

where ϕ_2 represents the phase fraction of opx. In the later discussion, ϕ_1 will be used to denote the phase fraction of olivine respectively. We also express bulk specific enthalpy with temperature and latent heat

$$\overline{H} = \phi_l h_l / c_p + \phi_s h_s / c_p + \phi_l L / c_p = \phi_l T + \phi_s T + \phi_l L / c_p = T + \phi_l L / c_p. \tag{1.49}$$

According to equations (1.47), the difference between melting point and eutectic point, which is defined as $\Delta T = T_m - T_e = T_{m0} - T_{e0}$, remains constant regardless of pressure change. At any given pressure, we proceed by dividing this ΔT to get a dimensionless formulation as

$$\frac{\overline{H}}{\Delta T} = \frac{T}{\Delta T} + \frac{\phi_l L}{c_p \Delta T}. \tag{1.50}$$

Let $\check{L} := \frac{L}{c_p \Delta T}$, $H := \frac{\overline{H}}{\Delta T}$ and $\check{T} := \frac{T}{\Delta T}$. We write the dimensionless enthalpy temperature relationship as

$$H = \check{T} + \phi_l \check{L}. \tag{1.51}$$

1.3.3 Phase Diagram

Binary eutectic melting is usually highly-nonlinear. For simplicity, we adopt a geometric parameterization of the binary eutectic diagram that can be analytically evaluated. We implement a linear eutectic phase boundary to distinguish between different phase regions within the diagram.

The eutectic composition of opx is set as $C_e \approx 0.25$. For illustration only, we plot bulk composition against a renormalized dimensionless temperature $\check{T} - T_e/\Delta T$, where $\check{T}_m - T_e/\Delta T = 1$ and $\check{T}_e - T_e/\Delta T = 0$.

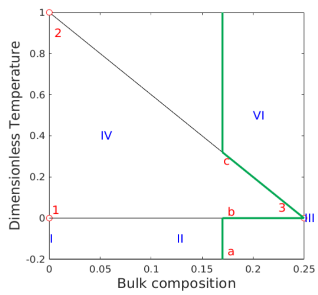

Figure 1.3

Figure 1.4: Phase diagram when the composition of opx is smaller than the eutectic compositon.

We identify the following 6 regions with different phase assemblages:

I	Single-phase sub-solidus	: Olivine
II	Two-phase sub-solidus	: Olivine + Orthopyroxene
III	Three-phase eutectic	: Olivine + Orthopyroxene + melt
IV	Two-phase super-eutectic	: Olivine + melt
V	Two-phase super-eutectic	: Orthopyroxene + melt
VI	Single-phase super-liquidus	: melt

In a partial melting system, we assume the bulk composition of orthopyroxene (opx) remains below the eutectic composition. Consequently, we focus in the part of the phase diagram 1.2, where the bulk composition is less than the eutectic composition, which is illustrated in the Figure 1.3.

The melting in eutectic system is illustrated with the green line in the Figure 1.3. Now imagining a system containing 17% opx starts melting. Starting with all solid phase at Point a, the system temperature increase linearly in response to the enthalpy rise. When the temperature reaches the eutectic temperature, at Point b, both components of the system starts melting with a fixed ratio of 25% opx and 75% olivine, which is the eutectic composition. Once all the solid phase opx is exhausted, the system leaves the eutectic points and solid olivine starts melting. The system climbs up along the line from Point 3 to Point c. After the remaining solid olivine melted, the temperature of the system will continue to rise linearly in response to enthalpy input. It is worth noting that we can reverse the process to obtain a path of equilibrium crystallization.

1.3.4 Phase Split Calculation

A detailed description of the phase-splitting calculations is provided for the regions I, II, III, and IV in this chapter. Regions V and VI are not considered in this study, as we assume that the opx composition remains below the eutectic composition and that only partial melting occurs within the system. We draw a Enthalpy-Composition diagram including the effect of latent heat in the Figure 1.5. Again, we plot against the renormalized enthalpy $H - T_e/\Delta T$ for simplicity. The individual concentration and temperature are determined using equations (1.48) and (1.51). Follow the prevous discussion, we denote the phase fraction of the olivine as ϕ_1 and the phase fraction of the opx as ϕ_2, such that $\phi_1 + \phi_2 = \phi_s$.

Region I. Pure olivine in the solid phase. The melting of pure olivine does not follow eutectic melting path. The conditions to be in the region I are

$$C = 0, \quad H \leq \check{T}_m, \tag{1.52}$$

and we can determine the phase fractions and the temperature as

$$\phi_1 = 1, \quad \phi_2 = \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{1.53}$$

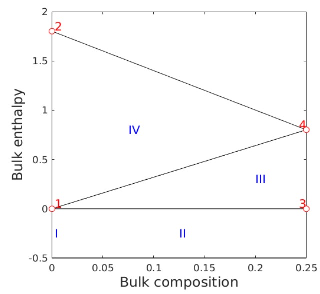

Figure 1.5: Enthalpy-Composition diagram

We can determine partial derivatives of the temperature with respect to bulk enthalpy and bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.54}$$

Region II. Sub-solidus: Olivine and opx coexist in the solid phase. In this phase region, temperature hasn't reached eutectic melting temperature. The conditions to be in the region II are

$$C \neq 0, \quad H \leq \check{T}_e, \tag{1.55}$$

and we can determine the phase fractions and the temperature as

$$\phi_2 = C, \quad \phi_1 = 1 - \phi_2, \quad \phi_l = 0 \quad \text{and} \quad \check{T} = H. \tag{1.56}$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 1 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.57}$$

Region III. Two-phase eutectic: The system starts melting at eutectic temperature with a fixed liquid concentration. Both solid phase olivine and orthopyroxene and melt coexist at the same time. Temperature remains at eutectic point until the opx exhausts. Concentration of the opx in the melt is also fixed equal to eutectic composition. The conditions to be in the region III are

$$C \leq C_e, \quad \check{T}_e < H \leq \text{line}_{1-4}, \tag{1.58}$$

where line_{1-4} is defined as $H(C) = \check{T}_e + \check{L}C/C_e$. We can determine the phase fractions and the current temperature as

$$\phi_l = \frac{H - \check{T}_e}{\check{L}}, \quad \phi_2 = C - \phi_l C_e, \quad \phi_1 = 1 - \phi_l - \phi_2 \quad \text{and} \quad \check{T} = \check{T}_e. \tag{1.59}$$

We can determine the partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\frac{\partial \check{T}}{\partial H} = 0 \quad \text{and} \quad \frac{\partial \check{T}}{\partial C} = 0. \tag{1.60}$$

Region IV. Super-eutectic two phase region: All solid phase opx has been exhausted. The concentration of opx in liquid c_l is now approximated with a linear function. The conditions to be in the region IV are

$$C \leq C_e, \quad \text{line}_{1-4} < H \leq \text{line}_{2-4}, \tag{1.61}$$

where line_{2-4} is defined as $H(C) = H_m - c_l/C_e$, and we can determine phase fractions as

$$\phi_2 = 0, \quad \phi_l = \frac{C}{c_l(\check{T})} \quad \text{and} \quad \phi_1 = 1 - \phi_l. \tag{1.62}$$

The corresponding temperature can be determined by solving the following equation

$$F(\check{T}) = H - \check{T} - \frac{C}{c_l(\check{T})}\check{L} = 0. \tag{1.63}$$

Let $c_l = C_e(\check{T}_m - \check{T})$, we can obtain a quadratic equation about temperature as

$$\check{T}^2 - (H + \check{T}_m)\check{T} + \check{T}_m H - CL/C_e = 0, \tag{1.64}$$

which can be solved with standard quadratic root finding formula. We omit one of the roots that exceeds melting temperature,

$$\begin{aligned} \check{T} &= \frac{H + \check{T}_m - \sqrt{(H + \check{T}_m)^2 - 4(\check{T}_m H - C\check{L}/C_e)}}{2} \\ &= \frac{H + \check{T}_m - \sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}{2}. \end{aligned} \tag{1.65}$$

We can determine partial derivatives of the temperature with respect to the bulk enthalpy and the bulk composition as

$$\begin{aligned}\frac{\partial \check{T}}{\partial H} &= \frac{1}{2}\left(1 - \frac{H - \check{T}_m}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}\right)\\ \frac{\partial \check{T}}{\partial C} &= -\frac{\check{L}/C_e}{\sqrt{(H - \check{T}_m)^2 + 4C\check{L}/C_e}}.\end{aligned} \tag{1.66}$$

1.4 Nondimensionalization

Compaction and melting are two most important aspects in mantle dynamics. We nondimensionlize the scaled formulation with a characteristic length scale related to compaction length and scaled characteristic Darcy speed of the melt,

$$l_0 = \Big(\frac{k_0\mu_s}{\mu_l}\Big)^{1/2} \quad \text{and} \quad u_0 = \frac{k_0\rho_r g}{\mu_l}. \tag{1.67}$$

We define corresponding characteristic time scale as $t_0 = l_0/u_0$, and characeristic pressure scale as $p_0 = \rho g l_0$. A set of nondimensional variables are defined as

$$\check{\mathbf{v}}_r = \frac{\mathbf{v}_r}{u_0}, \quad \check{\mathbf{v}}_s = \frac{\mathbf{v}_s}{u_0}, \quad \check{q} = \frac{q}{p_0}, \quad \check{q}_l = \frac{q_l}{p_0}. \tag{1.68}$$

The resulting dimensionless mechanics and transport equations are

$$\check{\mathbf{v}}_r + \phi^{1+\Theta}\nabla(\phi^{-1/2}\check{q}_l) = 0, \tag{1.69}$$

$$\phi^{-1/2}\nabla\cdot(\phi^{1+\Theta}\check{\mathbf{v}}_r) + \frac{1}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.70}$$

$$\nabla\check{q} - \nabla\cdot\check{\tau}(\check{\mathbf{v}}_s) = -(1-\phi)\mathbf{n}, \tag{1.71}$$

$$\nabla\cdot\check{\mathbf{v}}_s - \frac{\phi^{1/2}}{1-\phi}(\check{q}_l - \phi^{1/2}\check{q}) = 0, \tag{1.72}$$

where dimensionless deviatoric stress of the solid phase is $\check{\tau} = 2(1-\phi)(\frac{1}{2}(\nabla\check{\mathbf{v}}_s + \nabla\check{\mathbf{v}}_s^T) - 1/3\nabla\cdot\check{\mathbf{v}}_s\mathbf{I})$, which maintains the same form as (1.20). $\mathbf{n}$ is the unit direction vector pointing downwards. Now we rewrite transport equations with dimensionless

Quantity	Value	Units
ρ	3000	$kg \cdot m^{-3}$
ρ_r	600	$kg \cdot m^{-3}$
α_ρ	3×10^{-5}	K^{-1}
c_p	1200	$J \cdot kg^{-1} \cdot K^{-1}$
L	5×10^5	$J \cdot kg^{-1}$
μ_l	1	$\text{Pascal} \cdot s$
μ_s	10^{19}	$\text{Pascal} \cdot s$
k_0	10^{-8}	m^2
T_{e0}	1480	K
T_{m0}	2000	K
ν	6.5×10^6	$\text{Pascal} \cdot K^{-1}$

Table 1.1: Flow Mechanics and Thermaldynamics Parameters

variables as

$$\begin{aligned} \frac{\partial C}{\partial \check{t}} + \nabla \cdot (\check{\mathbf{v}}_{\text{eff}} C - \frac{\phi \mathcal{D}_l}{u_0 l_0} \nabla c_l) = 0, \\ \frac{\partial H}{\partial \check{t}} + \nabla \cdot (\check{\check{\mathbf{v}}} \check{T}) - \check{L} \nabla \cdot (\phi_s \check{\mathbf{v}}_s) - \frac{\kappa}{u_0 l_0} \nabla^2 \check{T} - \frac{\alpha_\rho l_0}{c_p} \check{T} \check{\check{\mathbf{v}}} \cdot \mathbf{g} = 0. \end{aligned} \tag{1.73}$$

Parameters used in the flow mechanics and thermaldynamic computations are listed in Table 1.1,

Chapter 2: Numerical Methods

We introduce a locally mass conserving numerical framework. We solve the static mechanics system with mixed finite element method, which is adjusted to be locally mass conservative. We solve the transport system with finite volume scheme, which is a mass conservative scheme.

2.1 Quadrilateral Mesh

We consider a bounded computational domain $\Omega \subset \mathbb{R}^2$. Here, $\partial\Omega$ denotes the Lipschitz-continuous boundary of the domain. Let $\mathcal{T}_h$ be a conforming mesh of quadrilaterals covering Ω. The quadrilateral $E \in \mathcal{T}_h$ is referred as element in finite element context, while referred as cell in finite volume context. Each quadrilateral E is assumed to be strictly convex, not degenerate into a triangle, line or point. For ease of implementation, we use a logically rectangular mesh, where each element(cell) is indexed using Cartesian coordinates $\{i, j\}$. We show a general quadrilateral element(cell) in Figure 2.1. The edges of element are denoted by e_i, $i \in \{0, 1, 2, 3\}$ in the counterclockwise direction. The vertices are represented as $v_i = e_i \cap e_{i+1}$ in the sense of modulo. Here, $\boldsymbol{\nu}_i$ and $\boldsymbol{\tau}_i$ are used to represent the unit normal and the unit tangent vector respectively.

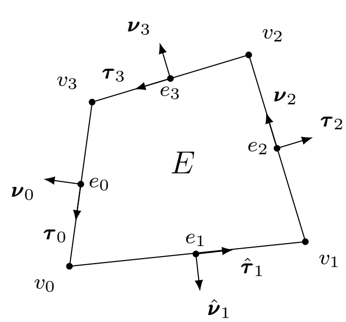

Figure 2.1: An exhibition of a general non-degenerate quadrilateral element(cell).

2.2 Shape Functions

2.3 Stokes Problem

We briefly describe solving static Stokes equation with mixed finite element scheme. We begin with strong form of the Stokes problem,

$$\begin{aligned} -\Delta \mathbf{u} + \nabla p &= \mathbf{f}, \\ \nabla \cdot \mathbf{u} &= 0. \end{aligned} \tag{2.1}$$

We derive weak form

$$\begin{aligned} (\nabla \mathbf{u}, \nabla \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{f}, \mathbf{v})_\Omega + \langle \tau^T \nabla \mathbf{u} \nu, \mathbf{v} \cdot \tau \rangle_{\Gamma_i}, \\ &\quad + \langle \nu^T \nabla \mathbf{u} \nu - p_0, \mathbf{v} \cdot \nu \rangle_{\Gamma_i} - \langle \nabla \mathbf{u}_0, \nabla \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.2}$$

Let stress $\mathbf{s} \cdot \tau = \tau^T \nabla \mathbf{u} \nu$ and $\mathbf{s} \cdot \nu = \nu^T \nabla \mathbf{u} \nu - p_0$, we have traction boundary condition term defined on Γ_i which is $< \mathbf{s}, \mathbf{v} >_{\Gamma_i}$. We employ a test space $\{\mathbf{v} \in H^1(\Omega) : \mathbf{v} = 0$ on $\Gamma_u\}$ here,

$$\begin{aligned} (\nabla \mathbf{u}_h, \nabla \mathbf{v}_h)_{\Omega_h} - (p_h, \nabla \cdot \mathbf{v}_h)_{\Omega_h} &= (\mathbf{f}, \mathbf{v}_h)_{\Omega_h} + \langle \mathbf{s}, \mathbf{v}_h \rangle_{\Gamma_{i,h}} - \langle \nabla \mathbf{u}_{0,h}, \nabla \mathbf{v}_h \rangle_{\Gamma_{u,h}}, \\ (\nabla \cdot \mathbf{u}, \omega) &= 0. \end{aligned} \tag{2.3}$$

We can separate normal and tangent part of stress on the boundary easily when physical boundary align with x-y axis. In the situation when a rectangular computational domain is used, we can mix stress and dirichlet boundary condition even further.

2.3.1 H^1 conforming discretization

2.4 Darcy Problem

$$\begin{aligned} \mathbf{u} + \nabla p &= \mathbf{g} \\ \nabla \cdot \mathbf{u} &= 0 \end{aligned} \tag{2.4}$$

weak form

$$\begin{aligned} (\mathbf{u}, \mathbf{v})_\Omega - (p, \nabla \cdot \mathbf{v})_\Omega &= (\mathbf{g}, \mathbf{v})_\Omega - \langle p_0, v_n \rangle_{\Gamma_i} - \langle \mathbf{u}_0, \mathbf{v} \rangle_{\Gamma_u} \\ (\nabla \cdot \mathbf{u}, \omega) &= 0 \end{aligned} \tag{2.5}$$

Test space $\{\mathbf{v} \in H(\mathrm{div}, \Omega) : v_n = 0 \text{ on } \Gamma_u\}$.

2.4.1 $H(\mathbf{div})$ Conforming discretization

2.5 Coupled Degenerate System

scaled nondimensionalized weak form let $\check{\sigma} = \mathcal{D}\mathbf{v} - \frac{1}{3}\nabla \cdot \mathbf{v}\mathrm{I}$

$$\begin{aligned}
(2\phi_s \check{\sigma}(\mathbf{v}_{s,h}), \nabla \Psi_s) - (\check{q}_h, \nabla \cdot \Psi_s) &= -(\phi_s \check{\mathbf{f}}, \Psi_s) \\
(\nabla \cdot \check{\mathbf{v}}_{s,h}, \omega) + \left(\frac{\hat{\phi}_f}{\phi_s}\check{q}_h, \omega\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_{f,h}, \omega\right) &= 0 \\
(\check{\mathbf{v}}_{r,h}, \Psi_r) - (\check{q}_{f,h}, \hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r)) &= 0 \\
(\hat{\phi}_f^{-1/2}\nabla \cdot (\phi_f^{1+\Theta}\check{\mathbf{v}}_r), \omega_f) + \left(\frac{1}{\phi_s}\check{q}_{f,h}, \omega_f\right) - \left(\frac{\hat{\phi}_f^{1/2}}{\phi_s}\check{q}_h, \omega_f\right) &= 0
\end{aligned} \tag{2.6}$$

2.5.1 Degenerate Saddle Point Problem

The linear system corresponding to (insert eqcite) has the form

$$\begin{bmatrix} \mathbf{A}_s & -\mathbf{B}_s & \mathbf{0} & \mathbf{0} \\ \mathbf{B}_s^T & \mathbf{C}_s & \mathbf{0} & \mathbf{K} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_d & -\mathbf{B}_d \\ \mathbf{0} & \mathbf{K} & \mathbf{B}_d^T & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{y}_s \\ \mathbf{x}_d \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{g}_s \\ \mathbf{f}_d \\ \mathbf{g}_d \end{bmatrix}, \tag{2.7}$$

wherein the solution represents the degrees of freem of primary variables with respect to the bases of the finite element spaces. (Insert details of calculation of each blocked matrix). We form a double saddle point system by symmetrically permutate original linear system,

$$\begin{bmatrix} \mathbf{A}_s & \mathbf{0} & -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d & \mathbf{0} & -\mathbf{B}_d \\ \mathbf{B}_s^T & \mathbf{0} & \mathbf{C}_s & \mathbf{K} \\ \mathbf{0} & \mathbf{B}_d^T & \mathbf{K} & \mathbf{C}_d \end{bmatrix} \begin{bmatrix} \mathbf{x}_s \\ \mathbf{x}_d \\ \mathbf{y}_s \\ \mathbf{y}_d \end{bmatrix} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \\ \mathbf{g}_s \\ \mathbf{g}_d \end{bmatrix}. \tag{2.8}$$

We group these block matrix to form a concise version,

$$\mathbf{A} = \begin{bmatrix} \mathbf{A}_s & \mathbf{0} \\ \mathbf{0} & \mathbf{A}_d \end{bmatrix}, \quad \mathbf{B} = \begin{bmatrix} -\mathbf{B}_s & \mathbf{0} \\ \mathbf{0} & -\mathbf{B}_d \end{bmatrix}, \quad \mathbf{C} = \begin{bmatrix} -\mathbf{C}_s & -\mathbf{K} \\ -\mathbf{K} & -\mathbf{C}_d \end{bmatrix}, \quad \mathbf{F} = \begin{bmatrix} \mathbf{f}_s \\ \mathbf{f}_d \end{bmatrix}, \quad \mathbf{G} = \begin{bmatrix} -\mathbf{g}_s \\ -\mathbf{g}_d \end{bmatrix}, \tag{2.9}$$

$$\begin{bmatrix} \mathbf{A} & \mathbf{B} \\ \mathbf{B}^T & \mathbf{C} \end{bmatrix} \begin{bmatrix} \mathbf{x} \\ \mathbf{y} \end{bmatrix} = \begin{bmatrix} \mathbf{F} \\ \mathbf{G} \end{bmatrix}, \tag{2.10}$$

where $\mathbf{A}$ is a positive definite matrix. However, $\mathbf{B}$ is usually a singular matrix when no melting region presents. We will present a modified Uzawa iteration algorithm to deal with this coupled degenerate singular system. Inexact Uzawa algorithm has been introduced in the previous literature Elman and Golub (1994) and James H. Bramble (1997),

Algorithm 1 Preconditioned Inexact Uzawa Iteration

1: Initialize $\mathbf{x}_0 = \mathbf{0}$ and $\mathbf{y}_0 = \mathbf{0}$
2: Update $\mathbf{x}_{i+1} = \mathbf{x}_i + \mathbf{A}^{-1}(\mathbf{F} - (\mathbf{A}\mathbf{x}_i + \mathbf{B}\mathbf{y}_i))$
3: Obtain $\mathbf{r} = \mathbf{B}^T\mathbf{x}_{i+1} + \mathbf{C}\mathbf{y}_i - \mathbf{G}$
4: Obtain $\tilde{\mathbf{y}} = \mathrm{argmin}_{\mathbf{x}\in V}||(\mathbf{B}^T\mathbf{A}^{-1}\mathbf{B} - \mathbf{C})^{-1}\tilde{\mathbf{y}} - \mathbf{r}||$
5: Update $\mathbf{y}_{i+1} = \mathbf{y}_i + \tilde{\mathbf{y}}$

2.6 Finite Volume Scheme

2.7 The Multi-level Weighted Essentially Non-Oscillatory (ML-WENO) Method

We develop a accurate, versatile, robust and efficient finite volume weighted essentially non oscillatory (WENO) techniques for solving energy and composition conservation laws in Arbogast et al. (2024).

2.7.1 Stencil Polynomial

Given a stencil $\mathcal{S} = \{E_0, E_2, \dots, E_k\}$ of the quasiuniform logically rectangular mesh. We can define an associated tensor product stencil polynomial $Q_{s,t} \in \mathbb{Q}_{(s,t)} := \mathbb{P}_{s-1}(x) \otimes \mathbb{P}_{t-1}(y)$, where s and t equal to the stencil sizes in the x- and y-directions, respectively. We scale our stencil polynomial individually as

$$Q_{s,t} = \sum_{p<s,q<t} c_{p,q}\Big(\frac{x - x_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^p\Big(\frac{y - y_{\mathcal{S}}}{h_{\mathcal{S}}}\Big)^q, \tag{2.11}$$

where $(x_{\mathcal{S}}, y_{\mathcal{S}})$ is the center point of the stencil $\Omega_{\mathcal{S}} = \cup_{E \in \mathcal{S}} \subset \mathbb{R}^2$, and $h_{\mathcal{S}} = \sqrt{\sum_k E_k/k}$ is the diameter of average cell size. Let $\xi = (x - x_{\mathcal{S}})/h_{\mathcal{S}}$ and $\eta = (y - y_{\mathcal{S}})/h_{\mathcal{S}}$ be the normalized varaibles, we can rewrite it into normalized version:

$$\hat{Q}_{s,t} = \sum_{p<s,q<t} c_{p,q} \xi^p \eta^q. \tag{2.12}$$

In finite volume scenerio, we are dealing with cell averaged value of the scalar transport variable

$$\overline{u}_E = \frac{1}{|E|} \int_E u(\mathbf{x}) d\mathbf{x} = \frac{1}{|E|} \int_E u(x, y) dx dy. \tag{2.13}$$

We can calculate coefficents $c_{p,q}$ by imposing a constraint that cell averaged value (2.13) is preserved through integral with stencil polynomial

$$\frac{1}{|E_k|} \int_{E_k} Q(\mathbf{x}) d\mathbf{x} = \overline{u}_{E_k}, \quad \forall E_k \in \mathcal{S}. \tag{2.14}$$

We will replace $\overline{u}_{E_k}$ with $\overline{u}_k$ in the following discussion for simplicity.

We can represent stencil polynomial with basis polynomials $Q_{s,t} = \sum_k \overline{u}_k \phi^k_{s,t}$, where basis polynomials can be calculated as

$$\frac{1}{|E_{k'}|} \int_{E_{k'}} \phi^k_{s,t}(\mathbf{x}) d\mathbf{x} = \frac{1}{|\hat{E}_{k'}|} \iint_{\hat{E}_{k'}} \hat{\phi}^k_{s,t}(\xi, \eta) d\xi d\eta = \begin{cases} 1 & k' = k \\ 0 & k' \neq k \end{cases} \quad \forall (E_{k'}, E_k) \in \mathcal{S}, \tag{2.15}$$

which would lead to a $(s \times t) \times (s \times t)$ non-singular linear system. We can now represent stencil polynomial with normalized basis polynomials as:

$$\hat{Q}_{s,t}(\xi, \eta) = \sum_k \overline{u}_k \hat{\phi}^k_{s,t}(\xi, \eta). \tag{2.16}$$

We define a linear projection operator $\pi_{\mathcal{S}}$ and let $\pi_{\mathcal{S}} u_{\mathcal{S}} = Q$ be a linear projection onto polynomial space W_p such that $\mathbb{Q}_{s,t} \subset W_p$. We estimate interpolation error with the Bramble-Hilbert Lemma Dupont and Scott (1980) and Bramble and Hilbert (1970) in $H_r(\Omega)$ space, where $r = s$ in x direction and $r = t$ in y-direction.

Lemma 2.1. *Let Ω_S be the domain covered by stencil $\mathcal{S}$ in $\mathbb{R}^2$ and let $u \in H^r(\Omega_{\mathcal{S}})$. We have:*

$$\|u - \pi_S u\|^2 \leq h^{2(r-1)} |u|^2_{H^r(\Omega_S)}. \tag{2.17}$$

Proof.

$$\|u - \pi_S u\| = \tag{2.18}$$

□

2.7.2 Stencil polynomial smoothness indicators

We define a smoothness indicator σ as a measurement of smoothness of $u(\mathbf{x})$ on the stencil S. The classic method from Jiang and Shu (1996) is defined by Sobolev seminorm as:

$$\sigma_P = \sum_{1 \leq |\alpha| \leq m} \int\int_{E_0} |E_0|^{|\alpha|-1} (\mathcal{D}^\alpha Q(x,y))^2 dx dy, \tag{2.19}$$

where $\alpha = (\alpha_1, \alpha_2)$ is a multi-index, $|\alpha| = \alpha_1 + \alpha_2$, and $\mathcal{D}$ is the derivative operator $\mathcal{D}^\alpha = \partial^{|\alpha|}/\partial x^{\alpha_1} \partial y^{\alpha_2}$. The Jiang-Shu smoothness indicator can be reformed with basis polynomials:

$$\begin{aligned}
\sigma_P &= \sum_{1 \leq |\alpha| \leq m} |E_0|^{|\alpha|-1} \iint_{E_0} (\mathcal{D}^\alpha Q(x,y))^2 dx dy \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} (\mathcal{D}^\alpha \hat{Q}(\xi,\eta))^2 d\xi d\eta \\
&= \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_0} \big(\sum_{ij} \overline{u}_{ij} \mathcal{D}^\alpha \hat{\phi}_{i,j}(\xi,\eta) \big)^2 d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'.j'} \sum_{1 \leq |\alpha| \leq m} \iint_{\hat{E}_{ij}} \mathcal{D}^\alpha \hat{\phi}_{ij}(\xi,\eta) \mathcal{D}^\alpha \hat{\phi}_{i'j'}(\xi,\eta) d\xi d\eta \\
&= \sum_{ij} \sum_{i'j'} \overline{u}_{ij} \overline{u}_{i'j'} \sigma_{ij}^{i'j'},
\end{aligned} \tag{2.20}$$

where $\sigma_{ij}^{i'j'}$ can be precomputed.

We proposed an alternative way of defining smoothness indicator in Huang et al. (2023).

$$\sigma_{\text{simp}} = \frac{1}{N_{\mathcal{S}} - 1} \sum_k (\overline{u}_k - \overline{u}_0)^2, \tag{2.21}$$

where $\overline{u}_0$ is the cell averaged solution in target cell. We define this simple smoothness indicator in order to reduce the compuational cost and to deal with distorted stencil on unstructured mesh. However, this simple smoothness indicator is not as robust as the classic Jiang-Shu smoothness indicator. Hence, we stick to the classic definition in the following discussion.

We observe that the smoothness indicator exhibit following convergence property that there exists some value $C > 0$ such that when $h \to 0^+$,

$$\sigma = \begin{cases} Ch^2 + \mathcal{O}(h^3) & \text{Smooth,} \\ \Theta(h) & \text{Not Smooth,} \end{cases} \tag{2.22}$$

where stencils are defined on quasi uniform mesh. This observation comes directly from the scaling property of Sobolev seminorm.

2.7.3 ML-WENO weighting

Given a target cell $E_{i,j}$, we can create a nonempty set of stencils $\mathcal{S}_l \ni E_{i,j}, l = 0, 1, \ldots, L$. The reconstruction of u on target cell Ei, j is a convex combination of stencil polynomials (2.16)

$$R(\mathbf{x}) = \sum_l \tilde{\omega}_l Q^l(\mathbf{x}), \quad \mathbf{x} \in E_{i,j}, \tag{2.23}$$

where $\sum_l \tilde{\omega}_l = 1$ is normalized nonlinear weights associated with stencils accordingly. We begin by arbitrarily chosing a set of linear weights $\{\omega_l\}$ and scale with stencil smoothness indicator as

$$\hat{\omega}_l = \frac{\omega_l}{(\sigma_l + \epsilon h_0^2)^{ar_l + n_l}}, \tag{2.24}$$

where $r_l = \max\{s, t\}$, $\epsilon = 1e - 4$ and $a > 0.5$ are picked as constants. The biasing factor $n_l = n(r_l)$ is used to biasing away further from constant level, whose smoothness

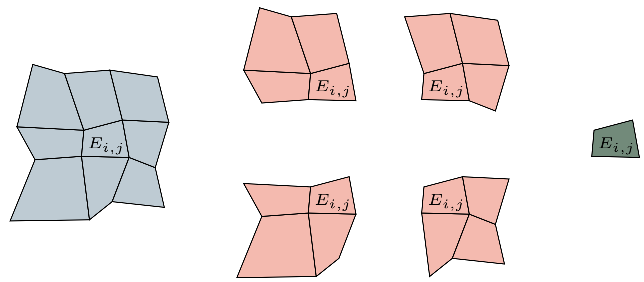

Figure 2.2: An example of (3,2,1) multi-level stencils combination on logically rectangular quadrilateral mesh.

indicator is always zero, when the solution is smooth. We pick

$$n(r_l) = n_l = \begin{cases} 1, & \text{if} \quad r_l = 1, \\ 3, & \text{if} \quad r_l = 2, \\ 4, & \text{if} \quad r_l \geq 3. \end{cases} \tag{2.25}$$

We then renormalize scaled weights

$$\tilde{\omega}_l = \frac{\hat{\omega}_l}{\sum_{l'} \hat{\omega}_{l'}}. \tag{2.26}$$

2.7.4 Boundary Treatment

2.8 Time Stepping

2.9 Boundary Condition Compatibility

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